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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's d, Pearson's r), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	For bulk RNA-sequencing, TARGET project gene expression data for high-risk neuroblastoma tumors were downloaded from https://treehousegenomics.soe.ucsc.edu/public-data/. The dataset used for analysis was "Tumor Compendium v11 Public PolyA (April 2020)" and only TARGET neuroblastoma ids were selected for analysis as they also had paired risk stratification data available. Gene level expression TPM data were inverse log-transformed. To compare with human normal tissue expression, gene level TPMs for diverse normal tissues were downloaded from the GTEx portal (https://gtexportal.org/home/datasets). Different subcategories of the same tissue type were collapsed to their respective tissue for ease of viewing. In total, we compared the gene expression of 30 unique tissue types (n= 15,989 total samples) from GTEx to 126 High-risk neuroblastoma tumors from the TARGET project. For single-cell RNA-sequencing, RDS files for neuroblastoma single-cell RNA-sequencing data were downloaded from (https://www.neuroblastomacellatlas.org/).
Data analysis	For bulk RNA-sequencing, boxplots were generated by ggplot and reshape2 R libraries in R 4.3.0. For single-cell RNA-sequencing, seurat objects were further scrutinized for certain gene expression patterns in already defined clusters. The clusters were defined previously using expression of lineage specific markers and further corroborated by running "quickMarkers" function in SoupX R package. FeaturePlot from the Seurat package was used to plot UMAPs. For all the other analysis, we used GraphPad Prism software. SingleR was run on the 2 neuroblastoma datasets using the Database Immune Cell Expression Data (DICE) that comprises 1561 bulk RNA-seq samples of sorted cell populations as a reference dataset. Prior to running SingleR, both datasets were filtered to remove high mitochondrial content and cells with low RNA count.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

High-risk neuroblastoma TARGET gene expression data were downloaded from (https://treehousegenomics.soe.ucsc.edu/public-data) and human normal tissue gene expression data were downloaded from the GTEx portal (https://gtexportal.org/home/datasets). RDS files for neuroblastoma single-cell RNA-sequencing data were downloaded from (https://www.neuroblastomacellatlas.org). All other data are available in the Article, Supplementary Information or Source Data files.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	A neuroblastoma tumor microarray (TMA) was constructed with duplicate punches from formalin-fixed paraffin-embedded human neuroblastoma tumors with diverse clinical features archived at the Children's Hospital of Philadelphia (CHOP). Fresh bone marrow samples from high-risk neuroblastoma patients with different clinical features were collected and banked at CHOP under the IRB-approved Protocol 10-007767 [CHOP Center for Childhood Cancer Research (CCCR) Biorepository and Registry].
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	For the TMA, A total of 64 unique neuroblastoma specimens were included, 31 of them defined as high-risk, 25 as intermediate-risk and 8 as low-risk. For BM samples, patient ages ranged from 2-14 years and patients were at initial diagnosis or had relapsed/refractory disease. All details are summarized in Supplementary Tables 1 and 2.
Recruitment	N/A
Ethics oversight	Patient samples were obtained under approved Institutional Review Board (IRB) protocols [tumor microarray (TMA); 18-015545 and cryopreserved bone marrow samples; 10-007767], and the study was conducted in accordance with the U.S. Common Rule. Healthy donors provided informed consent through the University of Pennsylvania Immunology Core.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

[x] Life sciences [ ] Behavioural & social sciences [ ] Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample size for mouse studies was based on our extensive experience with neuroblastoma pre-clinical models. For BiCE biodistribution, a minimum of 5 mice per group was sufficient to identify differences between BiCE tumor accumulation vs. healthy tissues. For NK-cell tumor retention, a minimum of 5 animals per group was sufficient to see specific intratumor retention of NK-cells in GPC2.CAR-GD2.BiCE T-cell-treated mice compared to GPC2.CAR-CD19.BiCE group. For efficacy studies, a minimum of 6 animals per group was sufficient to see antitumor efficacy of our targeted CARs compared to control CARs.
Data exclusions	No data were excluded from any study.
Replication	All in vitro experiments were done with at least 3 technical replicates. Functional T-cell and NK-cell-based in vitro assays were performed using at least 2 different individual donors as biological replicates.
Randomization	Mice with the appropriate enrollment tumor volumes were randomly assigned to a treatment cohort to ensure that each treatment group had comparable mean and median starting tumor volumes.
Blinding	There was no blinding.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems		
n/a	Involved in the study	
☐	☒	Antibodies
☐	☒	Eukaryotic cell lines
☒	☐	Palaeontology and archaeology
☐	☒	Animals and other organisms
☐	☒	Clinical data
☒	☐	Dual use research of concern
☒	☐	Plants

Methods		
n/a	Involved in the study	
☒	☐	ChIP-seq
☐	☒	Flow cytometry
☒	☐	MRI-based neuroimaging

Antibodies

Antibodies used

Western blot antibodies:
Rabbit HRP anti-6X His tag® antibody (ab1187; Abcam; 1:5000), rabbit CD3ζ (Cell Signaling; 88083; 1:800), rabbit CD3ζ phospho Y142 (Abcam; ab68235; 1:1000), rabbit Zap70 (Cell signaling; 3165S; 1:1000), rabbit phosho-Zap70 (Tyr319)/Syk (Tyr352) (Cell signaling; 2701; 1:800), and rabbit β-actin (Cell signaling; 4967S; 1:5000).

Flow cytometry antibodies:
PE anti-His-tag antibody (clone J095G46; 362603; Biolegend), anti-GD2 scFv (14g2a) idiotype (clone IA7; Creative Biolabs; FAMAB-0073-CN), PE-CD11b antibody (clone ICRF44; 301306; Biolegend), Ganglioside GD2 antibody (clone 14g2a; Biolegend), APC-MICA/MICB (clone 6D4; 320907; BioLegend), PE-ULBP1 (Clone # 170818; FAB1380P; R&D Systems), PE-B7-H6 (Clone # 875001; FAB7144P; R&D Systems), CD45-APC, PE or FITC (clone J33, Beckman Coulter), CD3-APC, PE or FITC (clone UCHT1; Beckman Coulter), CD19-PE (clone HIB19; 982402; BioLegend), CD16a-AlexaFluor®647 (Clone # 1001049; FAB4325R; R&D Systems), CCR7-FITC (clone G043H7; 353215; BioLegend), CD45RO-APC (clone UCHL1; 304210; BioLegend), PD1-APC (clone J105; eBioscience™), LAG3-PE (clone 3DS223H; eBioscience™), TIGIT-PE (Clone # 741182; FAB7898P; R&D Systems), CD69-FITC (clone FN50; 310903; BioLegend), CD107a-PE (clone H4A3; 328607; BioLegend), CD56-PE/Cyanine7 (clone HCD56; 318318; Biolegend), CD8-Brilliant Violet 421™ (clone SK1; 344748; Biolegend), CD4-PerCP/Cyanine5.5 (clone OKT4; 317428; Biolegend), CD3-APC/Cyanine7 (clone SK7; 344818; Biolegend), HLA-DR Brilliant Violet 785™ (clone L243; 307642; Biolegend), CD11b-PE/Cyanine7 (clone ICRF44; 301322; Biolegend), CD14-APC/Cyanine7 (clone 63D3; 367108; Biolegend), CD15-PerCP/Cyanine5.5 (clone W6D3; 323020; Biolegend), CD68-BD Horizon™ BV421 (clone Y1/82A; 564943; BDBiosciences), CD45-Pacific Blue™ (clone HI30; 304029; Biolegend), Human Aminopeptidase N/CD13 Alexa Fluor® 405 (Clone # 986002; AB38152V, R&D Systems), CD56 (NCAM)-Alexa Fluor® 488 (clone MEM-188; 304611; Biolegend).

Immunohistochemistry:
Rabbit polyclonal anti-human CD3 (Agilent; A0452; 1:100), monoclonal anti-human CD16a (C-terminal; SP189; ab227665; Abcam; 1:100) and Glypican-2 antibody (1:500; F-5, sc-393824, Santa Cruz Biotechnology).

Validation

GPC2 cell surface staining was performed using the D3-GPC2-IgG1 antibody conjugated to PE previously validated in Raman et al (Cell Reports Medicine, 2021).

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)

Human neuroblastoma NB-SD, CHP-134, NB-EbC1, SMS-SAN, SH-EP, SKNFI, NB69, NB-LS, LAN-5, Kelly, SY5Y, SK-N-AS, SMS-SAN and control NALM6 cell lines were obtained from the CHOP cell line repository, and the NK92 wild-type cell line was obtained from the Wistar Institute (Chi Van Dang laboratory). HEK293T and NK92-GFP-CD16 176V cell lines were obtained from ATCC (Catalog No. CRL-3216 and PTA-8836, respectively).

Authentication

Short-tandem repeat profiling of cell lines was provided by ATCC upon purchase. CHOP and Wistar cell lines were regularly authenticated by short-tandem repeat profiling.

Mycoplasma contamination

All cell lines tested negative for mycoplasma contamination.

Commonly misidentified lines
(See ICLAC register)

No misidentified cell lines were used.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Six-week-old immunodeficient female NSG (NOD-scid IL2Rgammanull; 005557; Jackson Labs) and NSG-MHC I/II DKO (024216; Jackson Labs) were utilized in this work.
Wild animals	N/A
Reporting on sex	Female mice were used in all in vivo studies.
Field-collected samples	N/A
Ethics oversight	Animal experiments were conducted using protocols approved by the CHOP Institutional Animal Care and Use Committee (IACUC; Protocols #643 and #1464) with adherence to the NIH guide for the Care and Use of Laboratory Animals accredited by the Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJEguidelines for publication of clinical research and a completedCONSORT checklist must be included with all submissions.

Clinical trial registration	N/A
Study protocol	N/A
Data collection	N/A
Outcomes	N/A

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A

Flow Cytometry

Plots

Confirm that:

- [x] The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- [x] The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- [x] All plots are contour plots with outliers or pseudocolor plots.
- [x] A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Single cell suspensions from human primary immune cells, bone marrow PBMCs, CAR T-cells, tumor or control cell lines, and neuroblastoma patient-derived xenografts were stained with LIVE/DEADTM Fixable Violet Dead Cell Stain (Invitrogen) or Zombie Aqua Fixable Viability Kit (77143; Biolegend) to determine cell viability before staining with the corresponding primary antibodies.

Instrument	Flow cytometry experiments were run on a Beckman CytoFLEX S cytometer (Beckam Coulter) or BD LSRII Cytometer (BDBiosciences).
Software	All flow cytometry data was analyzed using FlowJo 10.8.1 software.
Cell population abundance	No cell sorting was performed.
Gating strategy	For all flow cytometry studies, all cells were identified using SSC-A vs FSC-A gating, singlets using FSC-H vs FSC-A, live cells on the negative population of SSC-A vs PB-450 (for LIVE/DEADTM Fixable Violet Dead Cell Stain) or SSC-A vs PB-450 (Zombie Aqua Fixable Viability Kit). Gating strategies for bone marrow studies are exemplified in Extended Data Figures. For CAR expression, untransduced T-cells stained with the corresponding labelled recombinant protein were used to determine the boundary of CAR positivity. For phagocytosis assays, macrophages cultured alone defined the boundary of GFP positivity compared to macrophages co-cultured with GFP-positive tumor cells and antibodies. For NK-cell activation assays, NK-cells cultured alone determined the boundary of CD69/CD107a expression compared to NK-cells cultured with neuroblastoma cells and antibodies. For Transwell assays, cells from each well were collected and stained with CD45-FITC and CD3-APC to quantify the total amount of tumor cells (CD45-/CD3-), T-cells (CD45+/CD3+) and NK-cells (CD45+/CD3-) present in each Transwell well (either top or bottom).

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.